Wednesday, 1 April 2026 9:55 AM

How to Select optional value for k?

→ Elbow method

Elbow method

It is a method used in k-mean to find optional values for k so that we get clearer cluster.

How it works?

① It starts with all possible values of k

$K = 2, 3, 4, 5, 6, 7, 8, 10$

② For every k value implement K-means and find WCSS

WCSS → within cluster sum of squares.

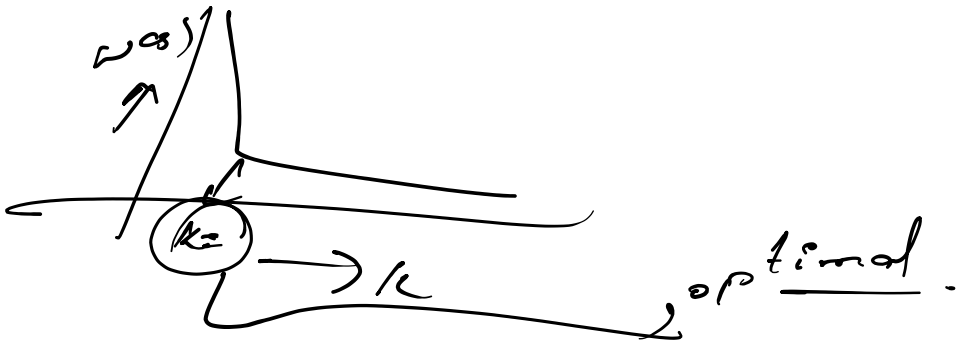
$$WCSS = \sum (x_i - c_i)^2$$

x_i → observation, c_i → Centroid.

Sum squared distance between observation and Centroid.

③ plot a graph for each k vs corresponding WCSS

and corresponding



optimal.